Quantum Mechanics II — Exercise Sheet 11

Scattering amplitudes and flux

Variant: Solutions

Exercise 1 (A downward step). A particle of energy $E > 0$ meets $V(x) = 0$ for $x < 0$ and $V(x) = -V_0$ for $x > 0$, with $V_0 > 0$. For unit incident amplitude from the left, match the wave and derivative at zero. (i) Find r, t, R and T. (ii) Explain why the transmitted amplitude need not be the square root of the transmission probability.

Solution. The left and right wave numbers are $k = \sqrt{2mE}/\hbar$ and $q = \sqrt{2m(E+V_0)}/\hbar$. Continuity gives $1 + r = t$ and $k(1 - r) = qt$. Solving gives

$$r = \frac{k-q}{k+q}, \qquad t = \frac{2k}{k+q}, \qquad R = \frac{(k-q)^2}{(k+q)^2}, \qquad T = \frac{4kq}{(k+q)^2}.$$

The current ratio supplies $T = (q/k)|t|^2$. The identity $(k-q)^2+4kq = (k+q)^2$ verifies $R+T = 1$. Different velocities on the two sides require the factor q/k.

Exercise 2 (Two delta barriers). Consider $V(x) = \alpha\delta(x) + \alpha\delta(x - d)$, with $\alpha > 0$ and $d > 0$. Take as given the composition formula

$$t_2(k) = \frac{t(k)^2 e^{\mathrm{i}kd}}{1 - r(k)^2 e^{2\mathrm{i}kd}}, \qquad t(k) = \frac{k}{k + \mathrm{i}\beta}, \quad r(k) = \frac{-\mathrm{i}\beta}{k + \mathrm{i}\beta}, \quad \beta = \frac{m\alpha}{\hbar^2}, \quad k > 0.$$

Derive $T_2 = |t_2|^2$ and the condition for perfect transmission. Show that no positive-energy pole occurs in the denominator.

Solution. Put $b = \beta/k$ and $\delta = kd$. The denominator after clearing t^2 is $(1 + \mathrm{i}b)^2 + b^2 e^{2\mathrm{i}\delta}$. Its squared modulus gives

$$T_2 = \frac{1}{1 + 4b^2(\cos\delta + b\sin\delta)^2}.$$

Perfect transmission occurs exactly when $\cos(kd) + (\beta/k)\sin(kd) = 0$. The squared denominator in the displayed expression is at least one. Alternatively, $|r(k)| < 1$ implies $|r(k)^2 e^{2\mathrm{i}kd}| < 1$ for real $k > 0$. Thus the resonance is a finite transmission maximum, not a pole on the positive real energy axis.

Exercise 3 (Resonances above a rectangular barrier). A barrier of height $V_0 > 0$ occupies $(0, \ell)$, and $E > V_0$. Let $k = \sqrt{2mE}/\hbar$ and $q = \sqrt{2m(E - V_0)}/\hbar$. Match $e^{\mathrm{i}kx} + re^{-\mathrm{i}kx}$ on the left, $Ae^{\mathrm{i}qx} + Be^{-\mathrm{i}qx}$ inside, and $te^{\mathrm{i}kx}$ on the right. Derive $T = |t|^2$ and find every energy of perfect transmission. Compute the limit as $E \downarrow V_0$ and decide whether the zero interior wave number is a perfect-transmission resonance.

Solution. The value and derivative at the right wall are $te^{\mathrm{i}k\ell}$ and $\mathrm{i}kte^{\mathrm{i}k\ell}$. Propagating the interior oscillatory solution to the left wall gives

$$1 + r = te^{\mathrm{i}k\ell}\left(\cos(q\ell) - \frac{\mathrm{i}k}{q}\sin(q\ell)\right),$$

$$\mathrm{i}k(1 - r) = te^{\mathrm{i}k\ell}\left(q\sin(q\ell) + \mathrm{i}k\cos(q\ell)\right).$$

Adding the first equation to the second divided by ik gives

$$t = \frac{e^{-ik\ell}}{\cos(q\ell) - i(k^2+q^2)\sin(q\ell)/(2kq)}, \qquad T = \left[1 + \frac{V_0^2 \sin^2(q\ell)}{4E(E-V_0)}\right]^{-1}.$$

Perfect transmission requires $q\ell = j\pi$ with $j \geq 1$, hence

$$E_j = V_0 + \frac{\hbar^2 \pi^2 j^2}{2m\ell^2}, \qquad j = 1, 2, \ldots.$$

At threshold $\sin(q\ell)/q \to \ell$, so

$$\lim_{E \downarrow V_0} T = \left[1 + \frac{mV_0\ell^2}{2\hbar^2}\right]^{-1} < 1.$$

The zero interior wave number is not a perfect-transmission resonance.

Exercise 4 (A packet averages probabilities). A normalized incoming momentum law is uniform on $[k_0, 2k_0]$ in wave number, where $k_0 > 0$. It scatters from a single delta barrier with $\beta = k_0$ and $T(k) = k^2/(k^2 + \beta^2)$. Compute the asymptotic transmission probability and compare it with $T(3k_0/2)$. For this delta model take as an additional given the packet identity

$$P_{\text{trans}} = \int_0^\infty T(k) w(k)\, dk,$$

where w is the normalized incoming wave-number density. The bounded-potential theorem in Lecture 11 is not being applied outside its hypotheses.

Solution. The supplied delta packet identity averages the transmission coefficient over the incoming law:

$$P_{\text{trans}} = \frac{1}{k_0} \int_{k_0}^{2k_0} \frac{k^2}{k^2 + k_0^2}\, dk = 1 - \arctan 2 + \frac{\pi}{4}.$$

The value at the midpoint is

$$T(3k_0/2) = \frac{9}{13}.$$

Numerically the average is about 0.67825, while the midpoint value is about 0.69231. Replacing a finite-width packet by its central wave number need not give the exact probability.